SOC 690S: Machine Learning in Causal Inference

Week 11: Panel Data and Differences-in-Differences

Wenhao Jiang
Department of Sociology, Fall 2025

Duke
UNIVERSITY

Week	Date	Topic	Problem sets Assign	Problem sets Due
1	Aug 26	Introduction: Motivation and Linear Regression		
2	Sep 2	Foundation: ML-based Regression Regularization		
3	Sep 9	Foundation: Non-Linear Machine Learning	1	
4	Sep 16	Foundation: Double Machine Learning		
5	Sep 23	Foundation: Directed Acyclic Graph (DAG)	2	1
6	Sep 29	Core: Matching, Propensity, Weighting, and Beyond		
7	Oct 7	Core: Instrumental Variable Estimation	3	2
8	Oct 14	*Fall break*		
9	Oct 21	**In-class midterm**		
10	Oct 28	Core: Regression Discontinuity Design		
11	Nov 4	Core: Panel Data and Difference-in-Difference		3
12	Nov 11	Advanced: Heterogeneous Treatment Effect I	4	
13	Nov 18†	Advanced: Heterogeneous Treatment Effect II		
14	Nov 25	Advanced: Unstructured Data Feature Engineering		4
	Dec 13	**final project due**		

† Justin Lucas Sola, Assistant Professor of Sociology and Data Science at UNC, will join us to give a guest talk on heterogeneous treatment effects and their applications.

Motivations for Using Panel Data

- So far, all models except for RDiT consider data that are *cross-sectional*
- This week, we move to a regime where there is another dimension of variation in time
- We consider strategies that use data with a time dimension to control for unobserved but fixed omitted variables
- These strategies punt on comparisons in levels while requiring the counterfactual *trend* behavior of treatment and control groups to be the same

Two-Way Fixed-Effects Model

Two-Way Fixed-Effects Model (TWFE)

- Given panel data (repeated observations on individuals), a natural estimator of the causal effect of D_i on Y_i is the fixed-effects (FE) model
- FE model assumes that

$$E[Y_{it}(0) \mid A_i, X_{it}, t, D_{it}] = E[Y_{it}(0) \mid A_i, X_{it}, t]$$

- Conditioning on A_i that is unobserved but time-invariant characteristics, D_i is as good as randomly assigned
- We assume linearity

$$E[Y_{it}(0) \mid A_i, X_{it}, t] = \alpha + \lambda_t + A_i'\gamma + X_{it}'\beta$$
$$E[Y_{it}(1) \mid A_i, X_{it}, t] = E[Y_{it}(0) \mid A_i, X_{it}, t] + \rho$$

Two-Way Fixed-Effects Model (TWFE)

- We therefore specify the linear regression estimating the effect of D_{it} on Y_{it} as

$$\begin{aligned} Y_{it} &= \alpha + \lambda_t + A_i'\gamma + X_{it}'\beta + \rho D_{it} + \epsilon_{it} \\ &= \alpha_i + \lambda_t + X_{it}'\beta + \rho D_{it} + \epsilon_{it} \end{aligned}$$

- where α_i is individual-specific intercepts, and the slope ρ is homogeneous across individuals
- FE is a *within estimator*, meaning that only within-individual variations are used in estimation

TWFE and Least-Squares Dummy Variable (LSDV) Model

- The FE model can be estimated by treating the individual effects as parameters
- This is called Least-Squares Dummy Variable (LSDV) Model
- The estimation can be computationally intense, and in most cases, we are not interested in individual-specific intercepts
- Algebraically, it is equivalent to estimation in deviations from means within individuals across time

$$\bar{Y}_i = \alpha_i + \bar{\lambda} + \bar{X}_i\beta + \rho\bar{D}_i + \bar{\epsilon}_i$$
$$Y_i - \bar{Y}_i = \lambda_t - \bar{\lambda} + (X_{it} - \bar{X}_i)\beta + \rho(D_{it} - \bar{D}_i) + \epsilon_{it} - \bar{\epsilon}_i$$

TWFE and First-Differences (FD) Model

- With two periods, deviation from means (TWFE) is algebraically the same as *differencing*, or first-Differences model

$$\Delta Y_{it} = \Delta\lambda_t + \rho\Delta D_{it} + \Delta X_{it}'\beta + \Delta\epsilon_{it}$$
$$\Delta Y_{it} = Y_{it} - Y_{it-1}$$

TWFE and First-Differences (FD) Model

- With more than two periods, *without* controls of X_{it}, and for a balanced panel, FE is algebraically the same as a weighted sum of *differencing* of all possible first-Differences models (Ishimaru 2025)

$$
\begin{aligned}
\hat{\beta}^{D}_{FE} &= \sum_{k=1}^{T-1} \widehat{w}^{D}_{k} \hat{\beta}^{D}_{FD,k} \\
\widehat{w}^{D}_{k} &= \left(\sum_{\ell=1}^{T-1} \sum_{i=1}^{N} \sum_{t=1}^{T-\ell} \Delta_{\ell} \widetilde{D}_{it} \Delta_{\ell} \widetilde{D}'_{it} \right)^{-1} \left(\sum_{i=1}^{N} \sum_{t=1}^{T-k} \Delta_{k} \widetilde{D}_{it} \Delta_{k} \widetilde{D}'_{it} \right)
\end{aligned}
$$

- $\Delta_k D_{it} = D_{it+k} - D_{it}$
- $\Delta_k \widetilde{D}_{it} = \Delta_k D_{it} - \frac{1}{N} \sum_{i=1}^{N} \Delta_k D_{it}$
- The FD periods with a larger total between-individual variation of changes in D_{it} are upward weighted

Two-Way Fixed-Effects (TWFE) Model

- Fixed-effects model does not have to rely on panel data
- Model where units that are nested within a higher structure, such as family or neighborhood, can include FE
- Ashenfelter and Krueger (1994) estimated returns to schooling using samples of twins, controlling for family fixed effects
- Since there are two twins from each family, this is the same as regressing Differencess in earnings within twin pairs on Differencess in schooling

TWFE, Differences-in-Differences, and Parallel Trend

- Indeed, TWFE is a *regression framework* that has the same regression specification and essence of the Differences-in-Differences (DiD)
  - Modern DiD typically refers to the binary-treatment, staggered adoption setup
  - Binary treatment that turns on at different times for different units, and once a unit becomes treated, it stays treated
  - TWFE deals with D_{it} that can be both discrete or continuous, and the change of D_{it} does not have to be staggered
- Similar to DiD, the estimate from TWFE is only a valid causal estimate if parallel trends hold
  - Units with different treatment intensities would have followed parallel trends in outcomes over time if treatment had not changed

TWFE and Lagged Dependent Variable (LDV) Model

- TWFE may be used to estimate the effect of (before and after) participation in a subsidized training program on wages (Dehejia and Wahba 1999; Lalonde 1986)
- However, the notion that the most important omitted variables are time invariant may not seem plausible
- It is likely that people looking to improve their labor market options have suffered some kind of setback
  - Training participants typically have earnings histories that exhibit a pre-program dip—a reverse selection problem
  - Past earnings is a time-varying confounding variable that cannot be absorbed in a time-invariant fixed-effects

TWFE and Lagged Dependent Variable (LDV) Model

- To control for such reverse selection, a Lagged Dependent Variable (LDV) Model may be specified

$$Y_{it} = \alpha + \theta Y_{i,t-h} + \lambda_t + \delta D_{it} + X_{it}'\beta + \epsilon_{it}$$

- To make it more general, $Y_{i,t-h}$ can be a vector including lagged earnings for multiple periods (Abadie, Diamond, and Hainmueller 2007)

TWFE and Lagged Dependent Variable (LDV) Model

- To control for such reverse selection, a Lagged Dependent Variable (LDV) Model may be specified

$$Y_{it} = \alpha + \theta Y_{i,t-h} + \lambda_t + \delta D_{it} + X_{it}'\beta + \epsilon_{it}$$

- To make it more general, $Y_{i,t-h}$ can be a vector including lagged earnings for multiple periods (Abadie, Diamond, and Hainmueller 2007)
- Applied researchers using panel data are often faced with the challenge of choosing between TWFE and LDV models
- One "solution" is to include both effects; but with small T, the estimate is both inconsistent and biased (Nickell 1981)

TWFE and Lagged Dependent Variable (LDV) Model

- To see the Nickell bias (1981), using differencing

$$\Delta Y_{it} = \theta \Delta Y_{it-1} + \Delta \lambda_t + \delta \Delta D_{it} + \Delta X'_{it} \beta + \Delta \epsilon_{it}$$

- The differencing residual $\Delta \epsilon_{it}$ and ΔY_{it-1} are both a function of ϵ_{it-1} and necessarily correlated

TWFE and Lagged Dependent Variable (LDV) Model

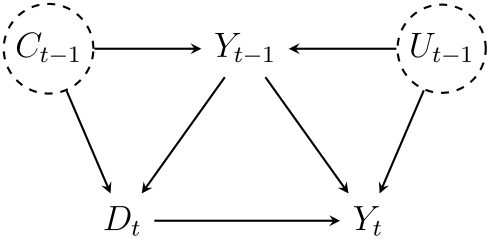
- To see the Nickell bias (1981), using differencing

$$\Delta Y_{it} = \theta \Delta Y_{it-1} + \Delta \lambda_t + \delta \Delta D_{it} + \Delta X_{it}' \beta + \Delta \epsilon_{it}$$

- The differencing residual $\Delta \epsilon_{it}$ and ΔY_{it-1} are both a function of ϵ_{it-1} and necessarily correlated
- The problem may be solved with strong assumptions
- For example, using Y_{it-2} or ΔY_{it-2} as an instrument for ΔY_{it-1} (Arellano and Bond 1991; for sociological equivalence using SEM, see Moral-Benito, Allison, and Williams 2019)
    - The first-stage is satisfied
    - But it requires that Y_{it-2} be uncorrelated with $\Delta \epsilon_{it}$, which may not be plausible if ϵ_i is serially correlated
    - Most people's earnings are highly correlated from one year to the next, so that past earnings are also likely to be correlated with $\Delta \epsilon_{it}$

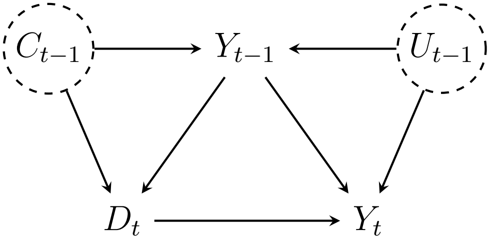
TWFE and Lagged Dependent Variable (LDV) Model

- The LDV is likely a *collider* (Dafoe 2018)



TWFE and Lagged Dependent Variable (LDV) Model

- The LDV is likely a *collider* (Dafoe 2018)



- Both TWFE and LDV rely on their respective assumptions
- The best practice is to specify both to check robustness of the results
- Indeed, if the assumption of LDV is correct while TWFE is mistakenly used, estimates of positive treatment effect will tend to be too big
- One may think of TWFE and LDV as bounding the causal effect of interest (MHE pp. 246)

Differences-in-Differences Model

Differences-in-Differences Model: Setup

- A particular case of TWFE is Differences-in-Differences (DiD), where time-invariance omitted variables may be captured by group-level fixed-effects
- Treatment is binary, the change of treatment is staggered, and there are always control units over time
- The canonical DiD is the simplest 2 by 2 case: one treatment time, one control group

Differences-in-Differences Model: Setup

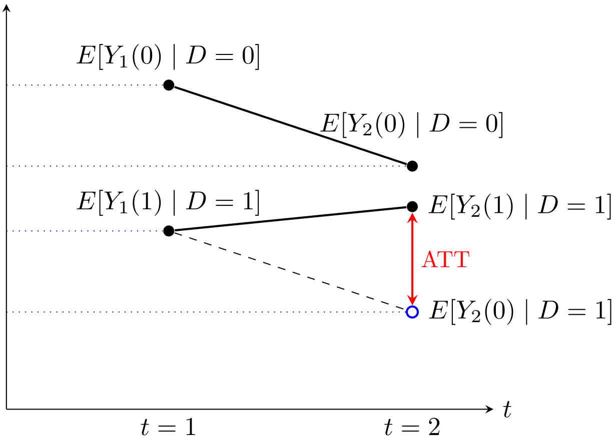
- In this classic paper, Card and Krueger (1994) studied the effect of raising minimum wage on employment
- On April 1, 1992, New Jersey raised the state minimum wage from $4.25 to $5.05, while just across the Delaware River, the minimum wage in eastern Pennsylvania stayed at $4.25
- The authors collected employment data at fast food restaurants in both NJ and PA in February 1992 and again in November 1992

Differences-in-Differences: Card and Krueger (1994)

Table: Average employment per store before and after the New Jersey minimum wage increase

Variable	*PA*	*NJ*	*NJ-PA*
FTE employment before	23.33	20.44	-2.89
	(1.35)	(0.51)	(1.44)
FTE employment after	21.17	21.03	-0.14
	(0.94)	(0.52)	(1.07)
Change in FTE employment	-2.16	0.59	2.76
	(1.25)	(0.54)	(1.36)

Differences-in-Differences: Setup



Parallel Trends: $E[Y_2(0) \,|\, D = 1] - E[Y_1(0) \,|\, D = 1] =$
$E[Y_2(0) \,|\, D = 0] - E[Y_1(0) \,|\, D = 0]$
No Anticipation: $E[Y_1(0) \,|\, D = 1] = E[Y_1(1) \,|\, D = 1]$

Differences-in-Differences: Setup

- We define potential outcomes $Y_t(d)$
- where $d \in \{0, 1\}$ denotes the treatment state in period $t = 2$
- Observed outcome in period t may then be represented as

$$Y_t = DY_t(1) + (1 - D)Y_t(0)$$

- We are left with missing data where we are unable to observe observations simultaneously in the treatment and control state

Differences-in-Differences: Setup

- Parallel trends and no anticipation assumption

$$E[Y_2(0) - Y_1(0) \mid D = 1] = E[Y_2(0) - Y_1(0) \mid D = 0]$$
$$E[Y_1(0) \mid D = 1] = E[Y_1(1) \mid D = 1]$$

- The parallel trends assumption is typically *functional form dependent*; it will generally not be the case that

$$E[g(Y_2(0)) - g(Y_1(0)) \mid D = 1] = E[g(Y_2(0)) - g(Y_1(0)) \mid D = 0]$$

- For example, parallel trends holding for wages does not imply that parallel trends holds for log(wages), and a DiD estimator based on log(wages) need not recover a causal effect (Roth and Sant'Anna 2023)
- Intuitively, this functional form dependence arises because parallel trends relies on latent confounding factors being additively separable, so that they are eliminated by the differencing operation

Differences-in-Differences: Setup

- Under the two assumptions, the treatment effect on the treated (ATT) can be estimated as

$$
\begin{aligned}
\alpha &= E[Y_2(1) - Y_2(0) \mid D = 1] \\
&= E[Y_2(1) \mid D = 1] - E[Y_2(0) \mid D = 1] \\
&= E[Y_2(1) \mid D = 1] - E[Y_2(0) - Y_1(0) \mid D = 1] - E[Y_1(0) \mid D = 1] \\
&= E[Y_2(1) \mid D = 1] - E[Y_2(0) - Y_1(0) \mid D = 0] - E[Y_1(1) \mid D = 1] \\
&= E[Y_2(1) - Y_2(1) \mid D = 1] - E[Y_2(0) - Y_1(0) \mid D = 0]
\end{aligned}
$$

Differences-in-Differences: Estimation

- Estimation of ATT in this canonical DiD in a finite sample is straightforward by considering four group means

$$\hat{\theta}_s(d) = \frac{\mathbb{E}[Y \cdot \mathbb{1}(D = d, t = s)]}{\mathbb{E}[\mathbb{1}(D = d, t = s)]}$$
$$\hat{\alpha} = (\hat{\theta}_2(1) - \hat{\theta}_1(1)) - (\hat{\theta}_2(0) - \hat{\theta}_1(0))$$

- Asymptotic properties under independence follow in a fashion similar to difference-in-mean estimators

Differences-in-Differences: Estimation

- Regression provides a numerically equivalent estimation of the ATT
- The regression formulation is especially convenient for obtaining standard errors under difference dependence assumptions, and allows the inclusion of covariates for *conditional* DiD

$$Y_{it} = \beta_0 + \beta_1 D_i + \beta_2 T_t + \alpha D_i \times T_t + \epsilon_{it}$$

where D_i denotes treatment or control group, T_t represents which period (1 or 2), and $D_i \times T_t = 0$ for control groups, and $D_i \times T_t = 1$ when the treatment group moves to period 2 and get treated

- This is equivalent to the TWFE specification where the key variable of interest is the actual assignment of treatment

Differences-in-Differences: Estimation

- The TWFE equivalence of the canonical 2 by 2 DiD is

$$Y_{it} = \mu_i + \lambda_t + \alpha D_{it} + \epsilon_{it}$$

where $D_{it} = 1$ if unit i has been treated by time t, 0 otherwise

Differences-in-Differences: Estimation

- The TWFE equivalence of the canonical 2 by 2 DiD is

$$Y_{it} = \mu_i + \lambda_t + \alpha D_{it} + \epsilon_{it}$$

  where $D_{it} = 1$ if unit i has been treated by time t, 0 otherwise
- Applied researchers have long assumed that the TWFE equivalence can be extended to a multi-period staggered setting analogous to that estimated by classic two-group two-period DiD, with $\hat{\alpha}$ capturing ATT across all treated observations and all post-treatment periods
- Careful consideration of the assumptions is required

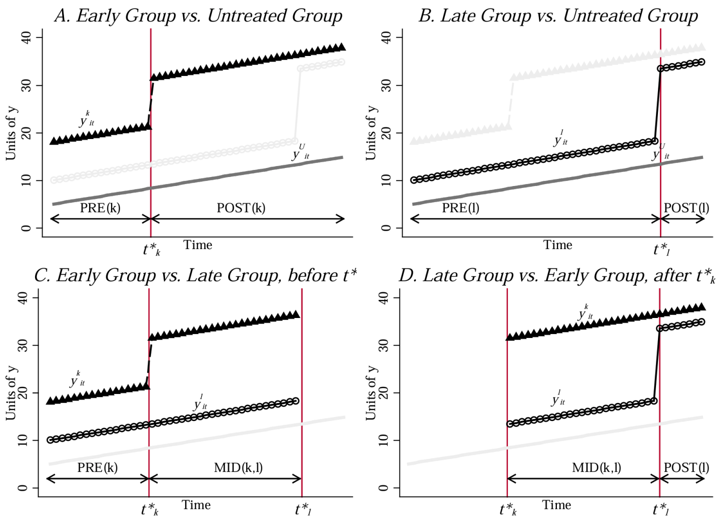
Differences-in-Differences: Goodman-Bacon (2021)

- Goodman-Bacon (2021) showed that the TWFE DiD estimator is a weighted average of many treatment effects of 2x2 DiDs
- Similar findings presented in other simultaneous work, such as de Chaisemartin and d'Haultfoeuille (2020)

Differences-in-Differences: Goodman-Bacon (2021)

- Consider the simplest staggered case, where the treatment is administered at two points in time as opposed to one
- This scenario yields two treatment groups, one group $G = 1$ which receives treatment at $t = e$ (the "early treated" group), and a second one $G = 2$ which receives treatment at $t = l$, where $l > e$ (the "late treated" group) as well as a "never treated" group that remains untreated throughout the panel
- Goodman-Bacon (2021) showed that this scenario can be broken down into four 2 by 2 DiD set-ups

Differences-in-Differences: Goodman-Bacon (2021)



TWFE as Weighted 2 by 2 DiDs: A Sketch

- TWFE estimate can be *numerically* shown to be equal to

$$
\begin{aligned}
\hat{\alpha}_{\text{TWFE}} &= \frac{\sum_i \sum_t y_{it} \tilde{D}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} \\
\tilde{D}_{it} &= (D_{it} - \bar{D}_i) - (\bar{D}_t - \bar{D}) \\
\bar{D} &= \frac{1}{NT} \sum_i \sum_t D_{it}
\end{aligned}
$$

- where the denominator is the total variance of the double-demeaned treatment indicator across all units and times

TWFE as Weighted 2 by 2 DiDs: A Sketch

- In the generalized scenario with G treatment groups, the TWFE estimate will be a weighted average of all possible 2 by 2s: early treated vs. late treated, late treated vs. early treated, and treated vs. untreated for each treatment group

$$\hat{\alpha}_{\text{TWFE}} = \sum_{g} \sum_{g'} w_{g,g'} \, \hat{\tau}_{g,g'}$$

- $\hat{\tau}_{g,g'}$ are the simple 2×2 DiD estimates comparing cohorts g and g' using all possible pre- and post-treatment windows
- $w_{g,g'}$ are the *positive* weights proportional to the contribution of that comparison to the overall variance of D_{it}

TWFE as Weighted 2 by 2 DiDs: A Sketch

- For example, for the comparison of cohort g (treated at time $t = g$) vs never-treated (∞), using pre $g-1$ and post $t \geq g$ windows, the 2×2 DiD ATT and weight is:

$$\hat{\tau}_{g,\infty} = \frac{1}{n_g} \sum_{i:G_i \in g} (Y_{it} - Y_{it-1}) - \frac{1}{n_\infty} \sum_{i:G_i \in \infty} (Y_{it} - Y_{it-1})$$

$$V_{g,\infty} = \sum_{i:G_i \in \{g,\infty\}} \sum_{t=g-1}^{t} \left(D_{it} - \bar{D}_i^{(g,\infty)} - \bar{D}_t^{(g,\infty)} + \bar{D}^{(g,\infty)} \right)^2$$

$$w_{g,\infty} = \frac{V_{g,\infty}}{\sum_{r,s} V_{r,s}}$$

TWFE as Weighted 2 by 2 DiDs: A Sketch

- The weight term has the exact same structure as the denominator of the TWFE main estimate $\hat{\alpha}_{\text{TWFE}}$

$$V_{g,\infty} = \sum_{i:G_i \in \{g,\infty\}} \sum_{t=g-1}^{t} \left(D_{it} - \bar{D}_i^{(g,\infty)} - \bar{D}_t^{(g,\infty)} + \bar{D}^{(g,\infty)} \right)^2$$

- The are both essentially the variance of double-demeaned treatment $\tilde{D}_{it}$; must be something deeper going on here

TWFE as Weighted 2 by 2 DiDs: A Sketch

- The weight term has the exact same structure as the denominator of the TWFE main estimate $\hat{\alpha}_{\text{TWFE}}$

$$V_{g,\infty} = \sum_{i:G_i \in \{g,\infty\}} \sum_{t=g-1}^{t} \left(D_{it} - \bar{D}_i^{(g,\infty)} - \bar{D}_t^{(g,\infty)} + \bar{D}^{(g,\infty)} \right)^2$$

- The are both essentially the variance of double-demeaned treatment $\tilde{D}_{it}$; must be something deeper going on here
- Remember in the anatomy of regression adjustment, we showed that regression coefficient is a weighted average of CATE, with weights proportional to the variance of D in covariate space X
- We have the double-demeaned treatment $\tilde{D}_{it}$; we do not have an explicit covariate space; instead, X is now implicitly the group $\{g, \infty\}$ in each 2 by 2 comparison to be "conditioned upon"

TWFE as Weighted 2 by 2 DiDs: A Sketch

- To make it more clear, the weights are proportional to the variance of $\tilde{D}_{it}$ for i in group $\{g, \infty\}$ across all possible 2 by 2 groups
- In this sense, the Goodman–Bacon decomposition is nothing new—it's a manifestation of the general OLS property that a single slope coefficient equals a variance-weighted average of the group-specific slopes
- Which cohorts g in any possible pairs of $\{g, g'\}$ over all possible 2 by 2 time intervals receive the largest weights?
  1. Naturally, cohorts where there are more observations
  2. Cohorts treated mid–panel with more balanced treatment and control distribution

TWFE as Weighted 2 by 2 DiDs, A Sketch

- Where are the “negative weights” that renewed our understanding of TWFE and DiD (Chaisemartin and d’Haultfoeuille 2020; Goodman-Bacon 2021)?
- The problem lies in the *forbidden comparison*, where late and early group may have different treatment effects in different time points
- “Negative weights only arise when average treatment effects vary over time” (Goodman-Bacon 2021)
- $\hat{\tau}_{g,g'}$ may be negative when early treated groups have larger treatment effect that accumulates over time, even if the treatment effect is positive

TWFE as Weighted 2 by 2 DiDs, A Sketch

- Where are the “negative weights” that renewed our understanding of TWFE and DiD (Chaisemartin and d’Haultfoeuille 2020; Goodman-Bacon 2021)?
- The problem lies in the *forbidden comparison*, where late and early group may have different treatment effects in different time points
- “Negative weights only arise when average treatment effects vary over time” (Goodman-Bacon 2021)
- $\hat{\tau}_{g,g'}$ may be negative when early treated groups have larger treatment effect that accumulates over time, even if the treatment effect is positive
- The “modern” DiD removes the pairs in *forbidden comparison*, and derives the re-weighted sum of all valid 2 by 2 ATTs; literature differs in how weights are derived
- The most intuitive weight is the OLS version of variance-based weight (after reweighting such that the sum equals to 1) (de Chaisemartin and D’Haultfoeuille 2020)

Conditional Differences-in-Differences

Conditional Differences-in-Differences

- The assumption that parallel trend holds for all units may be too strong
- Instead, it could be more plausible that parallel trends (and no anticipation) assumptions hold among units that are identical in terms of observed (time-invariant) characteristics

$$E[Y_2(0) - Y_1(0) \mid D = 1, X] = E[Y_2(0) - Y_1(0) \mid D = 0, X]$$
$$E[Y_1(0) \mid D = 1, X] = E[Y_1(1) \mid D = 1, X]$$

- We assume there is a *common support* of X between treatment and control group

$$\alpha(X) = E[Y_2(1) - Y_2(1) \mid D = 1, X] - E[Y_2(0) - Y_1(0) \mid D = 0, X]$$

and ATT is the weighted average of $\alpha(X)$ over the distribution of X in the treated group

Conditional Differences-in-Differences

- Remember since TWFE-based DiD is a weighted average of all possible 2 by 2s, treatment effects may differ by unit (group) and treatment time
- Therefore, in "modern" DiD approach that uses Double Machine Learning, it is natural to allow treatment effect heterogeneity across the support of X (and thus unit i)

Conditional Differences-in-Differences

- Remember since TWFE-based DiD is a weighted average of all possible 2 by 2s, treatment effects may differ by unit (group) and treatment time
- Therefore, in "modern" DiD approach that uses Double Machine Learning, it is natural to allow treatment effect heterogeneity across the support of X (and thus unit i)
- The most straightforward approach in a 2 by 2 pair is

$$\Delta Y = \alpha D + g(X) + g(X)'D + \Delta\epsilon$$

  which is essentially a Partial Linear Model (PLM) as we saw before in DML
- However, it still assumes linearity between treatment label and covariates space, with limited model generalizability
- Indeed, PLM may not uncover any sensible causal effect without strong functional form restrictions (Caetano and Callaway 2023)

Conditional Differences-in-Differences

- For flexibility and generalizability, the modern DML targeting conditional DiD is based on the *Interactive Regression Model* (IRM) that allows non-linear treatment effect heterogeneity across X
- More specifically, we assume the underlying DGP to be fully non-linear, with true nuisance function g_0 and m_0

$$\Delta Y = g_0(D, X) + \epsilon$$
$$D = m_0(X) + \mu$$

Conditional Differences-in-Differences and AIPW

- The IRM essentially has the same form as Augmented IPW—the canonical Doubly Robust estimand
- In DiD, we have ΔY instead of Y
- G-computation models conditional outcomes

$$g_0(D = d, X = x) = E[\Delta Y \mid D = d, X = x]$$

- The nuisance function m_0 models the treatment propensity conditional on X
- Therefore, as in AIPW, the ATE θ can be estimated as

$$\theta = E\left[g(1, X) - g(0, X) + \frac{D}{m(X)}\big(\Delta Y - g(1, X)\big) - \frac{1 - D}{1 - m(X)}\big(\Delta Y - g(0, X)\big)\right.$$

Conditional Differences-in-Differences and AIPW

- The AIPW-like estimand has several advantages
- It is doubly-robust: the estimand is unbiased and consistent as long as one of the nuisance functions is correctly specified
- It allows non-linear ML-ready estimation of the nuisance function
- Similar to AIPW (we explicitly proved it in Week 6), the estimand is *Neyman Orthogonal*, opening up the possibility of DML

$$\psi(W;\theta,g,m) =$$

$$g(1,X) - g(0,X) + \frac{D}{m(X)}\Big(\Delta Y - g(1,X)\Big) - \frac{1-D}{1-m(X)}\Big(\Delta Y - g(0,X)\Big) -$$

$$\frac{\partial}{\partial \eta} E[\psi(W;\theta_0,g,m)]\Big|_{\eta=\eta_0} = 0$$

Score Function for Conditional DiD

- As in AIPW, the above score function targets ATE
- In DiD, the estimand is ATT, where $g(1, X)$ does not need to be estimated; we need to estimate $g(0, X)$, the counterfactual of control
- For CATE $E[h(X)]$ under CIA, the CATT can be recovered by the rule of conditional probability

$$
\begin{aligned}
E[h(X) \mid D = 1] &= \frac{E[h(X)D]}{E[D]} \\
&= \frac{E[E[h(X)D \mid X]]}{E[D]} \\
&= \frac{E[h(X)m(X)]}{E[D]} \\
&= \frac{E[h(X)m(X)]}{p}
\end{aligned}
$$

Score Function for Conditional DiD

- Therefore, the moment condition for ATT is given by

$$
\begin{aligned}
&E[\psi(W;\theta,g,m)\mid D=1]= \\
&\quad \frac{1}{p}E\left[m(g_1-g_0)+D(\Delta Y-g_1)-\frac{m(1-D)}{1-m}(\Delta Y-g_0)-m\theta\right] \\
&\quad = E\left[\frac{D-m}{p(1-m)}(\Delta Y-g_0)-\frac{D}{p}\theta+\frac{m-D}{p}(g_1-g_0)\right] \\
&\quad = E\left[\frac{D-m}{p(1-m)}(\Delta Y-g_0)-\frac{D}{p}\theta\right]
\end{aligned}
$$

- Therefore, the score function for ATT in conditional DiD that is Neyman Orthogonal is (see pp. 458 and pp. 255 for IRM in CML)

$$
\frac{D-m}{p(1-m)}(\Delta Y-g_0)-\frac{D}{p}\theta
$$

DML for ATT in Conditional DiD

- Given the Neyman Orthogonial score, it is straightforward to implement DML to estimate ATT, where $\hat{\alpha}$ is $\sqrt{n}$-asymptotically normal
- Let $(W_i)_{i=1}^n = (Y_{1i}, Y_{2i}, D_i, X_i)_{i=1}^n$ be the observed data
- *Sample splitting.* Partition sample indices into random folds of approximately equal size:

$$\{1, \ldots, n\} = \bigcup_{k=1}^{K} I_k$$

For each $k = 1, \ldots, K$, compute estimators $\hat{p}_{[-k]}, \hat{g}_{[-k]}, \hat{m}_{[-k]}$, leaving out the kth fold of data

DML for ATT in Conditional DiD

- For each $i \in I_k$, let $k(i)$ denote the fold to which observation i belongs and define

$$\hat{\psi}(W_i;\alpha) = \frac{D_i - \hat{m}_{[k(i)]}(X_i)}{\hat{p}_{[k(i)]}\left(1 - \hat{m}_{[k(i)]}(X_i)\right)}\left(\Delta Y_i - \hat{g}_{[k(i)]}(0, X_i)\right) - \frac{D_i}{\hat{p}_{[k(i)]}}\alpha$$

- Compute $\hat{\alpha}$ as the solution to the estimating equation

$$\mathbb{E}_n[\hat{\psi}(W_i;\alpha)] = 0$$

- which yields

$$\hat{\alpha} = \frac{\frac{1}{n}\sum_{i=1}^{n} \frac{D_i - \hat{m}_{[k(i)]}(X_i)}{\hat{p}_{[k(i)]}\left(1 - \hat{m}_{[k(i)]}(X_i)\right)}\left(\Delta Y_i - \hat{g}_{[k(i)]}(0, X_i)\right)}{\frac{1}{n}\sum_{i=1}^{n} \frac{D_i}{\hat{p}_{[k(i)]}}}$$

DML for ATT in Conditional DiD

- Define
$$\hat{\phi}(W_i) = \frac{\hat{\psi}(W_i; \hat{\alpha})}{\frac{1}{n}\sum_{i=1}^{n} \frac{D_i}{\hat{p}_{[k(i)]}}}, \quad \hat{V} = \mathbb{E}_n[\hat{\phi}(W_i)^2]$$
- Based on the standard sandwich estimator of standard errors, construct SEs via $\sqrt{\hat{V}/n}$ and use standard normal critical values

Heuristic Test of Parallel Trend: Event Study

- We may test the parallel trend assumption by computing the "treatment effect" in pre-treatment period and see if there is significant "effect"
- Such non-zero placebo effects may suggest that treated and control group already differ in pre-treatment period
- The conventional approach based on TWFE simply specifies

$$Y_{it} = \mu_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbb{1}\{t - G_i = k\} + \epsilon_{it}$$

where G_i denotes the first period when unit i is treated, and $\mathbb{1}\{t - G_i = k\}$ is an indicator equal to one if unit i is k periods relative to its treatment adoption

Heuristic Test of Parallel Trend: Event Study

- The "pseudo" treatment effects in pre-treatment periods are compared with the omitted "baseline" period (typically $k = -1$)
- The coefficients β_k trace out the "event-study" or "dynamic treatment effects" over time, where coefficients on pre-treatment leads ($k < 0$) provide a heuristic test of the parallel trends assumption
- If the estimates of β_k for $k < 0$ are statistically indistinguishable from zero, it supports the assumption of parallel pre-treatment trends; otherwise, it suggests differential pre-trends

Heuristic Test of Parallel Trend: Event Study

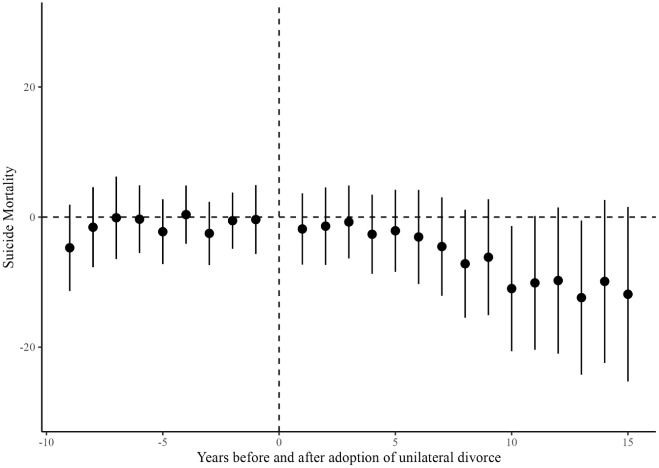
- The "pseudo" treatment effects in pre-treatment periods are compared with the omitted "baseline" period (typically $k = -1$)
- The coefficients β_k trace out the "event-study" or "dynamic treatment effects" over time, where coefficients on pre-treatment leads ($k < 0$) provide a heuristic test of the parallel trends assumption
- If the estimates of β_k for $k < 0$ are statistically indistinguishable from zero, it supports the assumption of parallel pre-treatment trends; otherwise, it suggests differential pre-trends
- However, as in the main DiD estimation, the *forbidden comparison problem* still exists, since TWFE event studies may use already-treated units as controls for newly treated ones
- "Modern" DiD estimators (e.g., Callaway and Sant'Anna, 2021; Sun and Abraham, 2021) address this issue by constructing event-study coefficients as weighted averages of valid 2 by 2 DiD comparisons

Stevenson and Wolfers (2006)

- In this paper, Stevenson and Wolfers study how changes in U.S. divorce laws—specifically, the shift to unilateral divorce—affected family outcomes such as suicide, domestic violence, and spousal homicide
- They use a differences-in-differences (DiD) framework exploiting staggered adoption of unilateral divorce laws across U.S. states and over time

Stevenson and Wolfers (2006)

Figure 3: TWFE Event Study Plot



Stevenson and Wolfers (2006)

Figure 4: Callaway & Sant'anna Method Event Study Plot